Crime4 Des

variable	dtype	label	missing	% missing
county	int16	county identifier	0	0.00
year	int8	81 to 87	0	0.00
crmrte	float32	crimes committed per person	0	0.00
prbarr	float32	'probability' of arrest	0	0.00
prbconv	float32	'probability' of conviction	0	0.00
prbpris	float32	'probability' of prison sentenc	0	0.00
avgsen	float32	avg. sentence, days	0	0.00
polpc	float32	police per capita	0	0.00
density	float32	people per sq. mile	0	0.00
taxpc	float32	tax revenue per capita	0	0.00
west	int8	=1 if in western N.C.	0	0.00
central	int8	=1 if in central N.C.	0	0.00
urban	int8	=1 if in SMSA	0	0.00
pctmin80	float32	perc. minority, 1980	0	0.00
wcon	float32	weekly wage, construction	0	0.00
wtuc	float32	wkly wge, trns, util, commun	0	0.00
wtrd	float32	wkly wge, whlesle, retail trade	0	0.00
wfir	float32	wkly wge, fin, ins, real est	0	0.00
wser	float32	wkly wge, service industry	0	0.00
wmfg	float32	wkly wge, manufacturing	0	0.00
wfed	float32	wkly wge, fed employees	0	0.00
wsta	float32	wkly wge, state employees	0	0.00
wloc	float32	wkly wge, local gov emps	0	0.00
mix	float32	offense mix: face-to-face/other	0	0.00
pctymle	float32	percent young male	0	0.00
d82	int8	=1 if year == 82	0	0.00
d83	int8	=1 if year == 83	0	0.00
d84	int8	=1 if year == 84	0	0.00
d85	int8	=1 if year == 85	0	0.00
d86	int8	=1 if year == 86	0	0.00
d87	int8	=1 if year == 87	0	0.00
lcrmrte	float32	log(crmrte)	0	0.00
lprbarr	float32	log(prbarr)	0	0.00
lprbconv	float32	log(prbconv)	0	0.00
lprbpris	float32	log(prbpris)	0	0.00
lavgsen	float32	log(avgsen)	0	0.00
lpolpc	float32	log(polpc)	0	0.00
ldensity	float32	log(density)	0	0.00
ltaxpc	float32	log(taxpc)	0	0.00
lwcon	float32	log(wcon)	0	0.00
lwtuc	float32	log(wtuc)	0	0.00
lwtrd	float32	log(wtrd)	0	0.00
lwfir	float32	log(wfir)	0	0.00
lwser	float32	log(wser)	0	0.00
lwmfg	float32	log(wmfg)	0	0.00
lwfed	float32	log(wfed)	0	0.00
lwsta	float32	log(wsta)	0	0.00
lwloc	float32	log(wloc)	0	0.00
lmix	float32	log(mix)	0	0.00
lpctymle	float32	log(pctymle)	0	0.00
lpctmin	float32	log(pctmin)	0	0.00
clcrmrte	float32	lcrmrte - lcrmrte[_n-1]	90	14.29
clprbarr	float32	lprbarr - lprbarr[_n-1]	90	14.29
clprbcon	float32	lprbconv - lprbconv[_n-1]	90	14.29
clprbpri	float32	lprbpri - lprbpri[t-1]	90	14.29
clavgsen	float32	lavgsen - lavgsen[t-1]	90	14.29
clpolpc	float32	lpolpc - lpolpc[t-1]	90	14.29
cltaxpc	float32	ltaxpc - ltaxpc[t-1]	90	14.29
clmix	float32	lmix - lmix[t-1]	90	14.29

Crime4 Summ

variable	N	mean	sd	min	max
county	630	100.6000	58.0363	1.0000	197.0000
year	630	84.0000	2.0016	81.0000	87.0000
crmrte	630	0.0316	0.0181	0.0018	0.1638
prbarr	630	0.3074	0.1712	0.0588	2.7500
prbconv	630	0.6886	1.6903	0.0684	37.0000
prbpris	630	0.4255	0.0872	0.1489	0.6786
avgsen	630	8.9545	2.6581	4.2200	25.8300
polpc	630	0.0019	0.0027	0.0005	0.0356
density	630	1.3861	1.4397	0.1977	8.8277
taxpc	630	30.2392	11.4547	14.3026	119.7615
west	630	0.2333	0.4233	0.0000	1.0000
central	630	0.3778	0.4852	0.0000	1.0000
urban	630	0.0889	0.2848	0.0000	1.0000
pctmin80	630	25.7129	16.9035	1.2837	64.3482
wcon	630	245.6661	121.9837	65.6216	2324.5984
wtuc	630	406.1028	266.5138	28.8577	3041.9580
wtrd	630	192.8231	88.4073	16.8738	2242.7473
wfir	630	272.0593	55.7681	3.5157	509.4655
wser	630	224.6704	104.8667	1.8438	2177.0681
wmfg	630	285.1701	82.3681	101.8300	646.8500
wfed	630	403.8959	63.0667	255.4000	597.9500
wsta	630	296.9074	53.4316	173.0200	548.0000
wloc	630	257.9762	41.3580	163.5900	388.0900
mix	630	0.1394	0.2198	0.0025	4.0000
pctymle	630	0.0890	0.0243	0.0622	0.2744
d82	630	0.1429	0.3502	0.0000	1.0000
d83	630	0.1429	0.3502	0.0000	1.0000
d84	630	0.1429	0.3502	0.0000	1.0000
d85	630	0.1429	0.3502	0.0000	1.0000
d86	630	0.1429	0.3502	0.0000	1.0000
d87	630	0.1429	0.3502	0.0000	1.0000
lcrmrte	630	-3.6092	0.5728	-6.3136	-1.8089
lprbarr	630	-1.2743	0.4159	-2.8332	1.0116
lprbconv	630	-0.6929	0.6096	-2.6827	3.6109
lprbpris	630	-0.8786	0.2305	-1.9042	-0.3878
lavgsen	630	2.1533	0.2737	1.4398	3.2515
lpolpc	630	-6.4906	0.5267	-7.6875	-3.3360
ldensity	630	-0.0159	0.7747	-1.6209	2.1779
ltaxpc	630	3.3564	0.3082	2.6604	4.7855
lwcon	630	5.4629	0.2482	4.1839	7.7513
lwtuc	630	5.9159	0.3702	3.3624	8.0203
lwtrd	630	5.2324	0.2144	2.8258	7.7155
lwfir	630	5.5794	0.2772	1.2572	6.2334
lwser	630	5.3646	0.3601	0.6118	7.6857
lwmfg	630	5.6152	0.2727	4.6233	6.4721
lwfed	630	5.9888	0.1588	5.5428	6.3935
lwsta	630	5.6778	0.1761	5.1534	6.3063
lwloc	630	5.5401	0.1597	5.0974	5.9612
lmix	630	-2.2342	0.6496	-6.0088	1.3863
lpctymle	630	-2.4430	0.1968	-2.7781	-1.2933
lpctmin	630	2.9134	0.9546	0.2497	4.1643
clcrmrte	540	0.0020	0.2029	-1.6501	1.2213
clprbarr	540	-0.0020	0.2749	-1.5805	1.2347
clprbcon	540	-0.0026	0.4837	-3.0456	2.1956
clprbpri	540	-0.0094	0.2757	-1.2356	1.1310
clavgsen	540	-0.0141	0.3223	-1.2788	1.2078
clpolpc	540	0.0129	0.2702	-3.1317	2.4234
cltaxpc	540	0.0773	0.1684	-0.7090	0.9322
clmix	540	-0.0037	0.4944	-2.9069	2.7287

Crime4 Iv Vs Ols Reg

IV vs OLS		
Variable	OLS	2SLS
const	0.030***	0.029***
	(0.001)	(0.003)
west	-0.015***	-0.015***
	(0.001)	(0.002)
central	-0.002*	-0.002
	(0.001)	(0.001)
urban	0.034***	0.034***
	(0.002)	(0.002)
polpc	1.291***	2.315
	(0.196)	(1.710)
N	630	630
R^2	0.454	0.430
Adj. R^2	0.451	0.427
F-stat	130.017	462.291
RMSE	0.013	0.014

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Crime4 Iv Vs Ols Diag

IV vs OLS					
Test	Stat	p	df	Dist	Reject
Wu-Hausman	0.376	0.540	1	F	No
H0: Regressors are exogenous (OLS consistent)					
Sargan	nan	nan	1	Chi2	No
H0: Overidentifying restrictions are valid					
Basmann	nan	nan	1	Chi2	No
H0: Overidentifying restrictions are valid					
Basmann F	nan	nan	1	F	No
H0: Overidentifying restrictions are valid					
Weak instr. F (polpc)	8.704	0.003	1, 625	F	Yes
H0: Instruments are irrelevant					

Crime4 Log Crime Ols Reg

Log Crime OLS				
Variable	Coef	Std Err	t	p
const	-2.350***	0.187	-12.560	0.000
	(0.187)			
lpolpc	0.154***	0.028	5.469	0.000
	(0.028)			
ldensity	0.515***	0.027	19.292	0.000
	(0.027)			
urban	-0.020	0.068	-0.290	0.772
	(0.068)			
west	-0.557***	0.038	-14.639	0.000
	(0.038)			
central	-0.322***	0.036	-8.993	0.000
	(0.036)			
N	630			
R^2	0.602			
Adj. R^2	0.599			
F-stat	188.784			
RMSE	0.363			

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Crime4 Fd Log Crime Reg

FD Log Crime				
Variable	Coef	Std Err	t	p
lpolpc	0.268***	0.030	8.869	0.000
	(0.030)			
ldensity	0.146	0.510	0.287	0.774
	(0.510)			
N	540			
R^2	0.128			
Adj. R^2	nan			
F-stat	39.333			
RMSE	0.175			

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Crime4 Fd Lead Test Reg

FD Lead Test				
Variable	Coef	Std Err	t	p
lpolpc	0.268***	0.030	8.869	0.000
	(0.030)			
ldensity	0.146	0.510	0.287	0.774
	(0.510)			
N	540			
R^2	0.128			
Adj. R^2	nan			
F-stat	39.333			
RMSE	0.175			

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Crime4 Fd Lead Test Diag

FD Lead Test					
Test	Stat	p	df	Dist	Reject
Lead test (strict exogeneity)	5.798	0.004	2, 89	F	Yes
H0: Coefficients on leads are jointly zero					

6.3 (B)

variable	dtype	label	missing	% missing
hhsize	float32	household size	0	0.00
povrat	float32	income to needs ratio	628	8.07
bmi	float32	own body mass index (wt for height measure)	1456	18.70
ageyrs	int8	age in years	0	0.00
female	int8	=1 if female	0	0.00
white	int8	=1 if white	0	0.00
black	int8	=1 if black	0	0.00
hisp	int8	=1 if hispanic	0	0.00
other	int8	=1 if other race	0	0.00
tvyest	float64	# hours of TV watched yesterday	1435	18.43
dadbmi	float32	dad's body mass index	1256	16.13
mombmi	float32	mom's body mass index	617	7.93
dadobese	float64	=1 if dad obese	1256	16.13
dadovwt	float64	=1 if dad overweight	1256	16.13
momobese	float64	=1 if mom obese	617	7.93
momovwt	float64	=1 if mom overweight	617	7.93
ovwtc	float64	=1 if overweight	1456	18.70
obesec	float64	=1 if obese	1456	18.70
pctcal	float32	calories consumed/2000	530	6.81

6.3 (C)

variable	N	mean	sd	min	max
obesec	6329	0.1343	0.3410	0.0000	1.0000
ageyrs	7785	9.4145	3.3664	5.0000	16.0000
female	7785	0.5070	0.5000	0.0000	1.0000
white	7785	0.3188	0.4660	0.0000	1.0000
black	7785	0.3105	0.4627	0.0000	1.0000
hisp	7785	0.3166	0.4652	0.0000	1.0000
tvyest	6350	3.1411	1.8018	0.0000	6.0000
povrat	7157	1.7303	1.3162	0.0000	7.5170

6.3 (D)

Variable	Coef	Std Err	t	p
const	-1.560***	0.131	-11.917	0.000
	(0.131)			
ageyrs	0.008	0.006	1.267	0.205
	(0.006)			
female	0.030	0.042	0.706	0.480
	(0.042)			
white	0.024	0.116	0.204	0.838
	(0.116)			
black	0.204*	0.112	1.828	0.068
	(0.112)			
hisp	0.261**	0.112	2.340	0.019
	(0.112)			
tvyest	0.057***	0.012	4.789	0.000
	(0.012)			
povrat	0.002	0.018	0.108	0.914
	(0.018)			
N	5770			
Pseudo R^2 (McFadden)	0.012			
$\text{Corr}^2(y, \hat{p})$	0.009			
Efron R^2	0.009			
AUC	0.582			
Log-likelihood	-2234.689			
LR stat	53.341			
LR p-value	0.000			
AIC	4485.379			
BIC	4538.662			
Brier score	0.114			
% correct (c=0.5)	86.707			
Balanced acc (c=0.5)	50.000			
% correct ($c = \bar{y}$)	52.288			
% correct (c* match $\bar{y}$)	78.267			
c* (match $\bar{y}$)	0.172			

Classification (c=0.5)	TP	TN	FP	FN
Counts	0	5003	0	767

Rate	TPR	TNR	PPV	NPV
Value	0.000	1.000	nan	0.867

Classification (c* match $\bar{y}$)	TP	TN	FP	FN
Counts	140	4376	627	627

Rate (c* match $\bar{y}$)	TPR	TNR	PPV	NPV
Value	0.183	0.875	0.183	0.875

Variable	AME	SE	p	MEM	SE	p
ageyrs	0.002	0.001	0.205	0.002	0.001	0.205
female	0.006	0.009	0.480	0.006	0.009	0.480
white	0.005	0.025	0.839	0.005	0.025	0.839
black	0.045	0.025	0.077	0.045	0.025	0.077
hisp	0.058	0.026	0.025	0.058	0.026	0.025
tvyest	0.012	0.003	0.000	0.012	0.002	0.000
povrat	0.000	0.004	0.914	0.000	0.004	0.914

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6.3 (E)

In the Probit model, the coefficient on `hisp` is positive and statistically significant ($p = 0.019$), so Hispanic children have a higher latent propensity to be obese, holding other covariates fixed. The average marginal effect is about 0.055 (SE 0.024, $p = 0.019$), implying roughly a 5.5 percentage-point higher probability of obesity on average.

6.3 (F)

Using the fitted Probit model, the predicted probability of obesity for a 16-year-old female white child who watched 3 hours of TV yesterday with income-to-needs ratio 4.56 is $\hat{p} = 0.114$ (about 11.4%).

6.3 (G)

If the same child watched no TV yesterday, the predicted probability falls to $\hat{p} = 0.084$ (8.4%). The drop is $\Delta\hat{p} = 0.0295$ (2.95 percentage points). Using a delta-method test for the difference gives SE 0.0062, $z = 4.78$, and $p \approx 1.8 \times 10^{-6}$, so the fall is statistically significant.

6.3 (H)

Joint test that `white`, `black`, and `hisp` are zero (Probit):

Test	Statistic	df	p-value
Wald (χ^2)	19.693	3	0.00020
LR (χ^2)	19.994	3	0.00017
Score/LM (χ^2)	19.738	3	0.00019

All three tests strongly reject the joint null at conventional levels.

6.3 (I)

Placeholder: potential sources of omitted-variable bias or reverse causality include:

- Unobserved diet quality / calorie intake beyond `pctcal`.
- Physical activity and sports participation.
- Parental BMI, genetics, and household health behaviors.
- Neighborhood environment (walkability, food access).
- Measurement error in TV hours or obesity status.
- Reverse causality: obese children may watch more TV because of lower activity or health constraints.

S7.3a Body

S7.3(A)

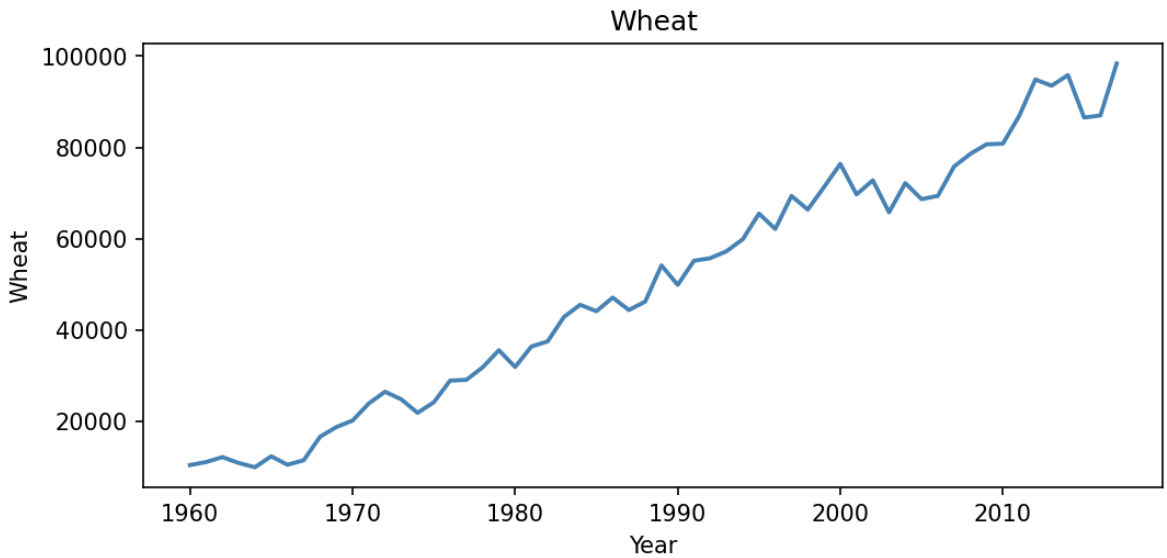

Figure 1: Time-series plot of wheat

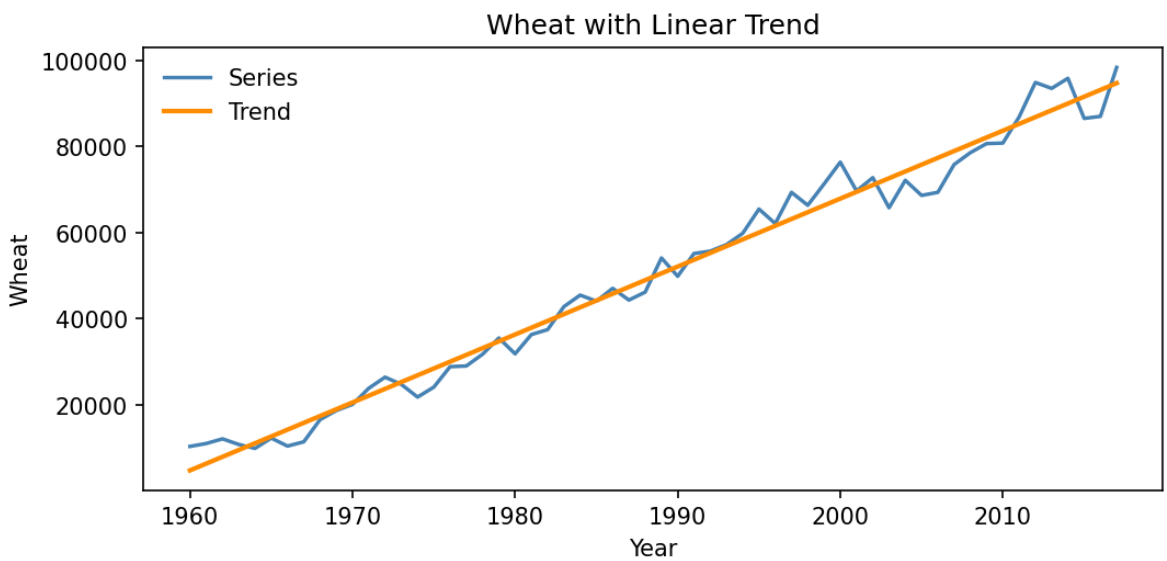

Figure 2: Wheat with a quadratic deterministic trend

The series shows a strong upward deterministic trend. A linear trend in time fits nearly as well as a quadratic one; AIC/BIC slightly favor the linear specification. A log-linear trend is also plausible if growth is proportional. Economically, the trend is consistent with technological improvements (Green Revolution), expanded irrigation, fertilizer use, improved seed varieties, and policy support that raised yields and production over time. (exponential growth) or a quadratic trend in time; the quadratic captures visible curvature in levels. Economically, the trend is consistent with technological improvements (Green Revolution), expanded irrigation, fertilizer use, improved seed varieties, and policy support that raised yields and production over time.

Method	RMSE	AIC	BIC	n
linear	3756.459	1123.420	1127.541	58
quadratic	3744.537	1125.051	1131.232	58
cubic	3744.701	1125.056	1131.237	58
log-linear	10548.084	-17.531	-13.411	58
log-quadratic	4225.527	-95.285	-89.103	58
hp	3516.936			58
moving_average	2345.683			58
lowess	3217.104			58

Table 1: Trend fit summary for wheat

S7.3b Body

S7.3(B)

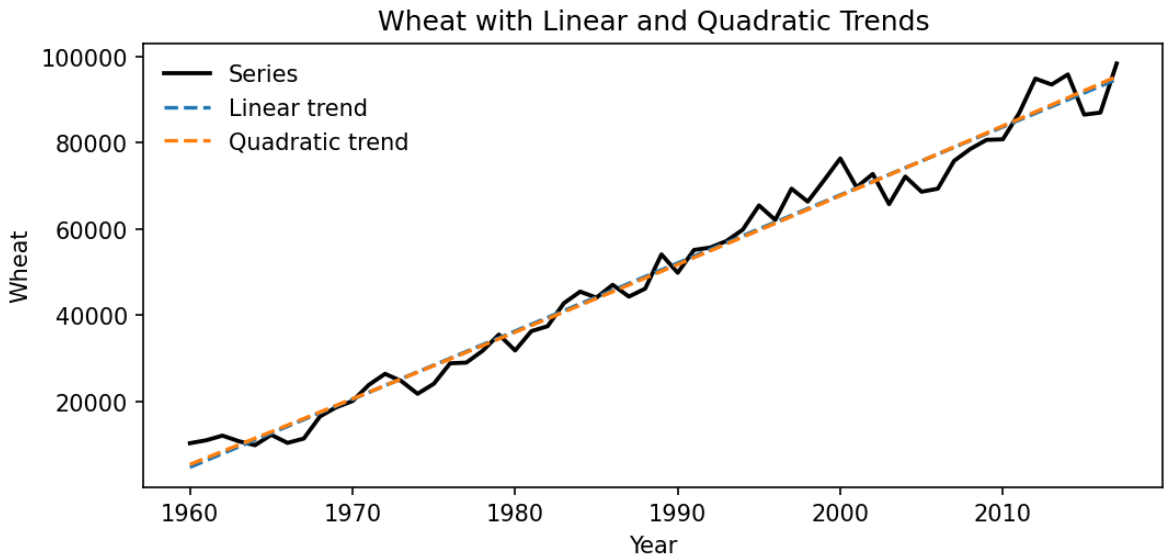

Figure 3: Wheat series with linear and quadratic trend fits (dashed)

AIC: linear=1123.420, quadratic=1125.051; BIC: linear=1127.541, quadratic=1131.232. AIC and BIC both prefer the linear trend. This makes intuitive sense because the quadratic term is small and adds little visible curvature; information criteria penalize the extra parameter when the fit gain is minimal.

S7.3c Body

S7.3(C)

Using the AIC-preferred linear trend, the 2018 forecast for wheat production is $\hat{y}_{2018} = 96309.4$ with a 95% confidence interval [94272.0, 98346.9]. This forecast assumes the deterministic trend is correctly specified, the error term is mean-zero with constant variance, and there are no structural breaks or policy/technology shocks between the sample and 2018.

S7.3d Body

S7.3(D)

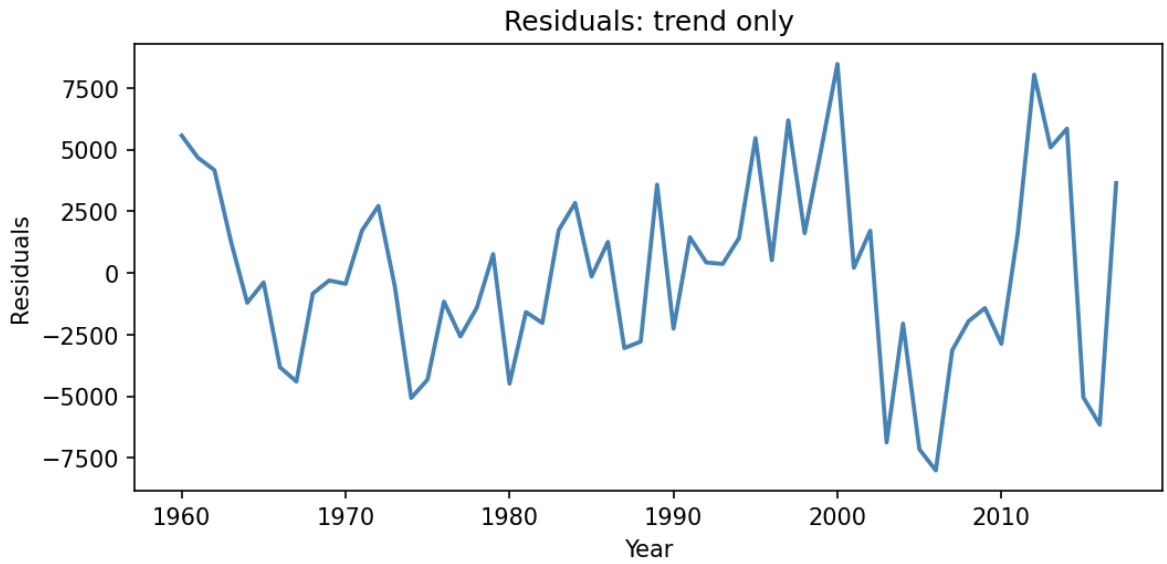

Figure 4: Residuals from trend-only model

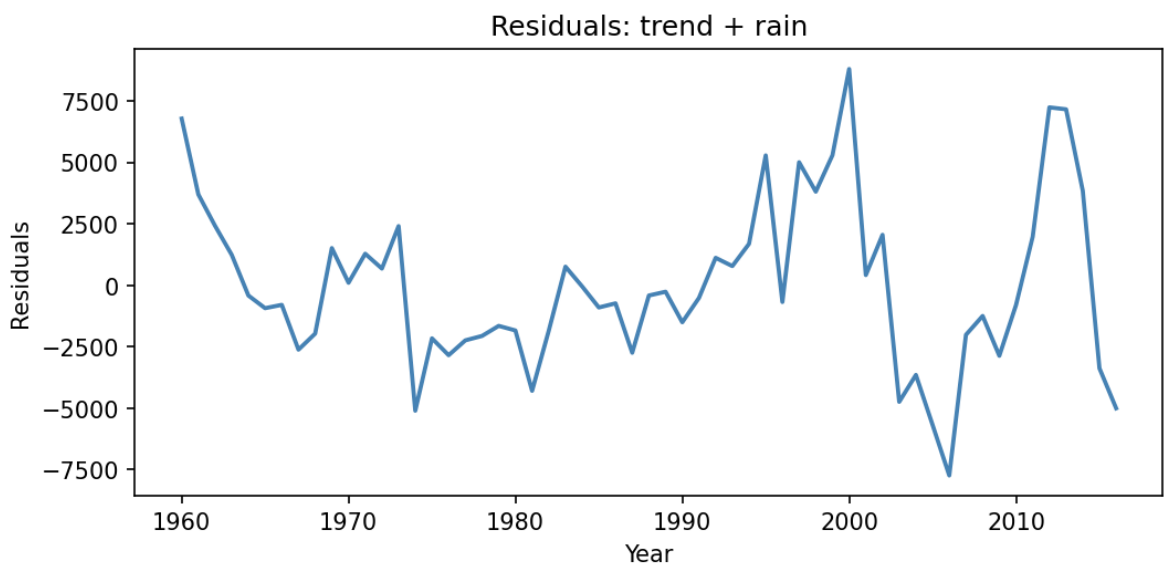

Figure 5: Residuals from trend + rain model

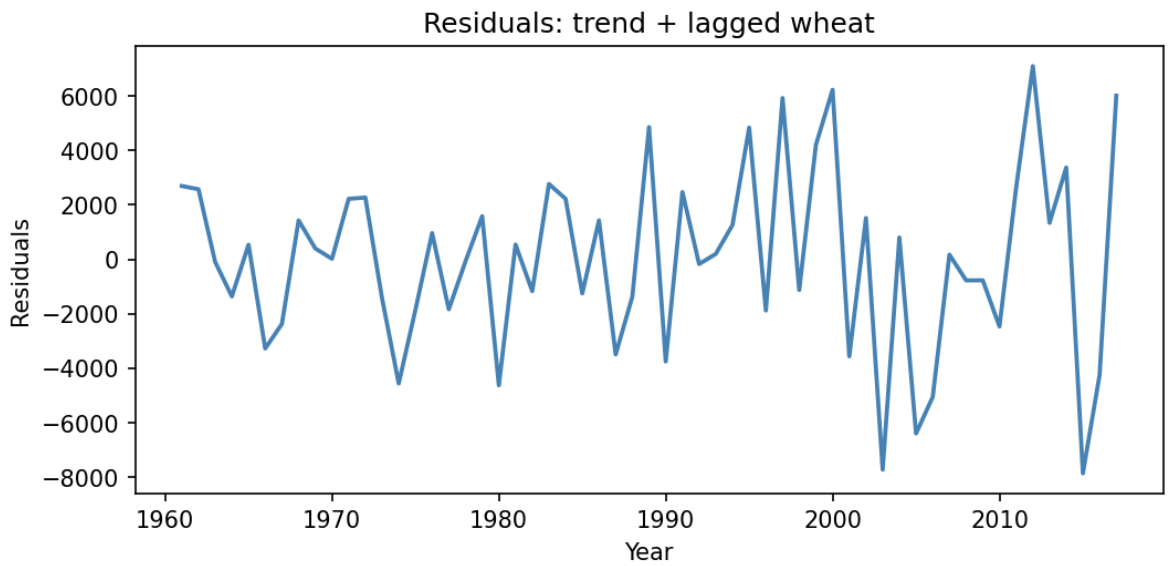

Figure 6: Residuals from trend + lagged wheat model

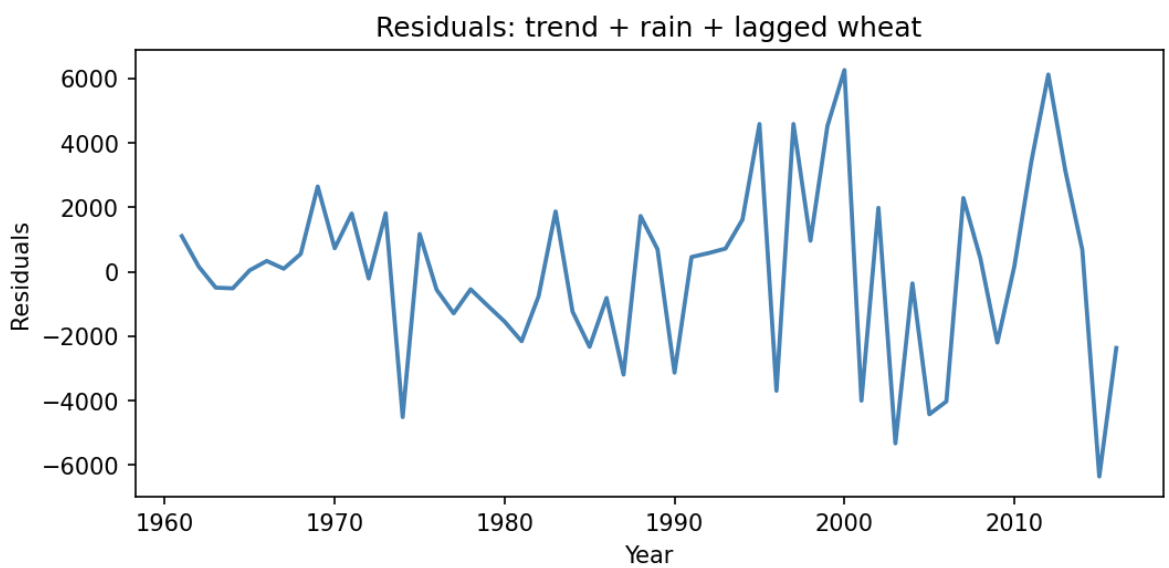

Figure 7: Residuals from trend + rain + lagged wheat model

Trend-only: $DW = 1.093$; Ljung–Box $Q(4) = 16.136$ ($p = 0.003$). Trend + rain: $DW = 0.766$; Ljung–Box $Q(4) = 27.315$ ($p = 0.000$). Trend + lagged wheat: $DW = 2.035$; Ljung–Box $Q(4) = 2.738$ ($p = 0.603$). Trend + rain + lagged wheat: $DW = 1.840$; Ljung–Box $Q(4) = 3.665$ ($p = 0.453$). Interpretation: adding lagged wheat substantially reduces serial correlation relative to trend-only or trend+rain, suggesting persistence in output that is not explained by rainfall alone. Rain may help explain levels but does not eliminate autocorrelation without a dynamic term.

S7.3e Body

S7.3(E)

We augment the trend model with rainfall to see whether it reduces residual dependence. Lag selection for Granger tests uses AIC with maxlags=4; selected lag = 1.

Series	Test	Stat	p	df	Dist	Reject
wheat (level)	ADF	0.216	0.973	4	ADF	No
wheat (level)	KPSS	1.249	0.010	4	KPSS	Yes
wheat (level)	PP	0.934	0.994	11	PP	No
wheat (diff)	ADF	-5.541	0.000	3	ADF	Yes
wheat (diff)	KPSS	0.106	0.100	5	KPSS	No
wheat (diff)	PP	-13.712	0.000	11	PP	Yes
rain (level)	ADF	0.115	0.967	7	ADF	No
rain (level)	KPSS	0.981	0.010	3	KPSS	Yes
rain (level)	PP	-8.703	0.000	11	PP	Yes
rain (diff)	ADF	-3.199	0.020	8	ADF	Yes
rain (diff)	KPSS	0.167	0.100	15	KPSS	No
rain (diff)	PP	-37.913	0.000	11	PP	Yes

Table 2: Stationarity tests for wheat and rain in levels and first differences

Test	Stat	p	df	Dist	Reject
Granger: rain $\rightarrow$ wheat	16.010	0.000	1, 53	F	Yes
Granger: wheat $\rightarrow$ rain	21.956	0.000	1, 53	F	Yes
Lead exogeneity (rain)	0.129	0.721	1, 50	F	No

Table 3: Exogeneity and Granger causality tests

Trend + rain model residual diagnostics: $DW = 0.766$; Ljung–Box $Q(4) = 27.315$ with $p = 0.000$. If residual serial correlation persists, the inference and forecast intervals from (c) are understated; a dynamic specification (e.g., AR errors, adding lagged wheat, or differencing) would be more appropriate.